

INFLUENTIAL NODE DETECTION IN WEIGHTED GRAPHS USING GRAPH CONVOLUTIONAL NETWORKS

A Project Report Submitted
in Partial Fulfilment of the Requirements
for the Degree of

Bachelor of Technology (Honours)

in

Computer Science with Specialization in AI and Data Science

by

Albin John
(Roll No. 2023BCD0005)



to

**DEPARTMENT OF COMPUTER SCIENCE WITH
SPECIALIZATION IN AI AND DATA SCIENCE
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
KOTTAYAM-686635, INDIA**

April 2026

DECLARATION

I, **Albin John** (Roll No: **2023BCD0005**), hereby declare that, this report entitled “**Influential Node Detection in Weighted Graphs using Graph Convolutional Networks**” submitted to Indian Institute of Information Technology Kottayam towards partial requirement of **Bachelor of Technology (Honours) in Computer Science with Specialization in AI and Data Science** is an original work carried out by me under the supervision of **Dr. Jisha Mariyam John** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold the academic ethics and honesty. Whenever an external information or statement or result is used then, that have been duly acknowledged and cited.

Kottayam-686635

Albin John

April 2026

CERTIFICATE

This is to certify that the work contained in this project report entitled “**Influential Node Detection in Weighted Graphs using Graph Convolutional Networks**” submitted by **Albin John** (Roll No: **2023BCD0005**) to the Indian Institute of Information Technology Kottayam towards partial requirement of **Bachelor of Technology (Honours)** in **Computer Science with Specialization in AI and Data Science** has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

Kottayam-686635

April 2026

(Dr. Jisha Mariyam John)

Project Supervisor

ABSTRACT

Identifying influential nodes in complex networks is a fundamental problem in network science. It has applications in viral marketing, epidemic control, and information diffusion. Recently, learning-based approaches such as Neural Local Graph Convolutional Networks (NLGCN) have shown impressive performance in predicting influential nodes in the graph.

In this project, we implement and analyze the NLGCN framework for identifying influential nodes in graphs. The model includes structural features such as Node Local Influence (NLI) and Node Global Influence (NGI) that are computed at multiple aggregation scales to construct a multi-channel representation for each node. A channel attention mechanism is implemented to learn the relative importance of these features, followed by convolutional layers to predict node influence scores.

Ground truth labels are generated using the SIR epidemic diffusion model , and performance is evaluated using Kendall Tau correlation. The implementation proves the effectiveness of learning-based approaches for influence ranking and provides a foundation for future enhancements.

Contents

List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Problem Statement	2
1.2 Objectives	2
1.3 Motivation	2
1.4 Organization of the Report	3
2 Literature Review	4
2.1 Traditional Centrality-Based Approaches	4
2.2 Deep Learning Approaches	5
2.3 Advanced and Hybrid Approaches	6
2.4 Survey of Existing Work	6
2.5 Research Gap	7
3 Methodology	9
3.1 Graph Representation	10
3.2 Local Node Influence (NLI)	10
3.3 Node Global Influence (NGI)	11
3.4 Multi-Scale Feature Construction	11
3.5 Neighbourhood Matrix Construction	12
3.6 Structural Channel Construction	12
3.7 Channel Tensor Formation	13
3.8 Channel Attention Mechanism	13

3.9	NLGCN Model Architecture	14
3.10	SIR-Based Label Generation	14
3.11	Model Training	15
3.12	Evaluation Metrics	15
3.12.1	Kendall Tau Correlation	15
3.12.2	Top- N Spread Evaluation	15
4	Experimental Results	17
4.1	Datasets	17
4.2	Experimental Setup	18
4.2.1	Framework and Hardware	18
4.2.2	Hyperparameters	18
4.2.3	Training Convergence	18
4.3	Baseline Methods	19
4.4	Comparison with Baseline Methods	19
4.4.1	Kendall Tau Correlation	19
5	Discussion	21
5.1	Interpretation of Results	21
5.1.1	Why NLGCN Outperforms Baseline Methods	21
5.1.2	Role of the Channel Attention Mechanism	22
5.2	Limitations	22
5.2.1	Scalability	22
5.2.2	Sensitivity to Hyperparameters	23
5.2.3	No Ablation Experiments	23
5.2.4	Static Graph Assumption	23
5.3	Comparison with Related Work	23
5.3.1	Heuristic Centrality Methods	23
5.3.2	GNN-Based Approaches	24
5.3.3	Survey Perspective	24
6	Conclusion and Future Work	25
6.1	Conclusion	25
6.2	Future Work	26

6.2.1	Extension to Weighted Graphs	26
6.2.2	Extension to Directed and Dynamic Graphs	27
6.2.3	Exploring Alternative Diffusion Models	27
6.2.4	Scalability to Large Graphs	28
Bibliography		29

List of Figures

3.1	Basic Framework of the NLGCN model showing (a) Node Feature Extraction, (b) Structural Channel Set Embedded with Feature Information, and (c) Model Prediction [3].	9
3.2	NLGCN model processing pipeline	14

List of Tables

2.1	Summary of existing approaches for influential node identification	7
4.1	Summary of Benchmark Datasets	17
4.2	Hyperparameter Configuration	18
4.3	Training Loss Convergence	19
4.4	Kendall Tau Correlation Comparison on Benchmark Datasets	19
4.5	NLGCN Percentage Improvement Over Baseline Methods	20

Chapter 1

Introduction

In real world systems, such as social networks, biological networks, transportation networks etc, some nodes play a very important role in the propagation of information, influence, or disease [4, 1, 2]. Identifying these *influential nodes* is key challenge in the network science and has significant practical applications [4]. For instance, in viral marketing, targeting a small set of highly influential users can lead to large scale information spread [4, 3], in epidemiology, immunizing key individuals can control the outbreak of a disease [1].

Traditional methods for finding the influential node depends on hand crafted centrality measures such as degree centrality, betweenness centrality [2], closeness centrality [2], and k -shell decomposition. While computationally efficient, these methods often fail to capture complex structural patterns, especially in weighted or heterogeneous graphs [5, 1]. Simulation-based approaches such as the Susceptible-Infected-Recovered (SIR) model provide more accurate influence estimates, but are computationally expensive and do not scale to large networks [3, 6].

Recent advances in graph neural networks (GNNs) and convolutional learning on graphs [9] open new possibilities for learning node representations that encode rich structural information [8, 6]. This project proposes a novel framework, the **Neural Local Graph Convolutional Network (NLGCN)**, which combines structural feature engineering with deep learning to predict influential nodes in weighted graphs efficiently and accurately [3].

1.1 Problem Statement

Given a weighted, undirected graph $G = (V, E)$, the goal is to assign an influence score to each node $v \in V$ such that nodes with higher scores are more likely to trigger large-scale information diffusion when selected as initial spreaders [4, 1]. The challenge lies in capturing both local neighbourhood structure and global network topology [5, 3] without resorting to expensive simulation procedures at inference time [6].

1.2 Objectives

The key objectives of this project are as follows:

- To design structural feature descriptors — Node Local Influence (NLI) and Node Global Influence (NGI), that capture multi-scale structural properties of nodes in weighted graphs [3, 5].
- To construct a multi-channel tensor representation for each node by embedding these features into neighbourhood subgraph matrices at three aggregation orders [3, 8].
- To design and train an attention-augmented convolutional neural network (NLGCN) that predicts node influence scores from the tensor representation [9].
- To generate ground truth influence labels using the SIR epidemic model and evaluate the proposed model using Kendall Tau correlation and Top- N spread metrics [3].

1.3 Motivation

The motivation for this work stems from two key observations. First, existing centrality-based methods are inadequate for capturing the nuanced influence dynamics in weighted and complex networks [1, 5, 2]. Second, while GNN-based approaches have shown promise [9, 6], they typically require attribute-rich graphs or focus on classification tasks rather than influence ranking [7, 8]. The proposed NLGCN framework solves this issue by combining domain-specific structural features with the representational power of convolutional learning [3], acquiring a fast, scalable, and accurate influential node detection.

1.4 Organization of the Report

The report is organized as follows. Chapter 2 presents a review of existing approaches for influential node detection, including traditional centrality-based methods, deep learning techniques, and hybrid approaches. Chapter 3 describes the proposed methodology in detail, covering graph representation, feature construction, model architecture, label generation, training procedure, and evaluation metrics. Chapter 4 shows the experimental results and comparative analysis. Chapter 5 discusses the findings and their implications. Chapter 6 concludes the report and outlines directions for future work.

Chapter 2

Literature Review

This chapter presents a review of existing approaches for identifying influential nodes in complex networks, covering traditional centrality-based methods, deep learning techniques, and hybrid approaches. A comparative summary of existing work is also provided, followed by a discussion of the research gaps that motivate the present study.

2.1 Traditional Centrality-Based Approaches

Traditional methods for identifying influential nodes rely primarily on handcrafted centrality measures derived from the topological properties of networks. These measures quantify the importance of a node based on its structural position within the graph [2, 1].

Farooq et al. [2] examined the detection of influential nodes using social network analysis based on standard network metrics. Their study evaluated degree centrality, betweenness centrality, and closeness centrality, demonstrating that these metrics capture different aspects of node importance. Degree centrality measures the number of direct connections a node has, betweenness centrality quantifies the extent to which a node lies on shortest paths between other nodes, and closeness centrality reflects the average distance from a node to all other nodes in the network. While computationally efficient, the authors noted that relying on any single metric provides an incomplete picture of a node's true influence.

Chen et al. [1] provided a systematic evaluation of various centrality measures and semi-local metrics for identifying influential nodes in complex networks. Their work high-

lighted that local measures such as degree centrality are computationally inexpensive but fail to capture the global positioning of nodes, whereas global measures like betweenness centrality offer better macroscopic insight at the cost of significantly higher computational complexity, rendering them unsuitable for large-scale networks.

Mukhtar et al. [5] conducted a comparative analysis of network centrality measures using ANOVA-based statistical testing. Their study demonstrated that different centrality metrics produce statistically distinct rankings for the same set of nodes, underscoring the inherent limitations of relying on a single topological descriptor for influence assessment and motivating the need for multi-feature approaches.

Thotakura et al. [7] applied centrality measures alongside community detection algorithms in the context of Alzheimer’s disease network analysis, demonstrating the applicability of traditional network science methods to biomedical domains. Their work showed that centrality measures can identify critical biomarkers and structural patterns in medical networks, further establishing the broad relevance of influential node identification.

2.2 Deep Learning Approaches

In recent years, deep learning has emerged as a powerful paradigm for learning from graph-structured data. Graph Convolutional Networks (GCNs) [9] have proven particularly effective for learning latent representations of nodes by recursively aggregating information from their local neighbourhoods. Unlike traditional methods that depend on predefined topological rules, GCNs can automatically learn optimal feature representations that encode both node-level attributes and graph-level structural information.

Zhang et al. [9] provided a comprehensive review of graph convolutional network architectures, covering spectral and spatial approaches to graph convolution. Their survey highlighted the ability of GCNs to capture multi-hop neighbourhood information through layer-wise propagation, making them highly suitable for tasks such as node classification, link prediction, and influence ranking in complex networks.

Hafiene et al. [4] presented an extensive survey on influential node detection in dynamic social networks. The survey reviewed methods ranging from static centrality measures to learning-based approaches and underscored the limitations of handcrafted features in dynamic and evolving networks. The authors highlighted that traditional top- k

ranking methods inevitably lose their effectiveness as graph structures grow in complexity and scale, motivating the adoption of data-driven learning techniques.

Patel and Singh [6] proposed a graph convolutional network for structural equivalent key node identification in complex networks (SE-GCN). Their model aims to preserve the structural role and equivalence of nodes while identifying key nodes. Although highly effective, the method relies on the availability of informative node features and can struggle with purely topological graphs where node attributes are absent or uninformative.

2.3 Advanced and Hybrid Approaches

To overcome the individual limitations of traditional and deep learning methods, several researchers have proposed hybrid architectures that integrate handcrafted topological features with neural network models.

Tu et al. [8] developed a hybrid CNN-GNN approach for identifying influential nodes in complex networks. Their method combines convolutional neural networks for capturing local compositional features with graph neural networks for encoding graph-level structural dependencies. This dual-pathway architecture demonstrated improved performance over both standalone CNN and GNN approaches, establishing the value of combining local and global structural information.

Guo et al. [3] proposed a key node identification method based on hybrid centrality combinations and graph neural networks. Their approach constructs multi-channel representations of nodes using localized and aggregated influence features — Node Local Influence (NLI) and Node Global Influence (NGI) — computed at multiple aggregation scales. These structural channels are processed through an attention mechanism followed by convolutional layers to predict node influence scores. This methodology serves as the primary foundation for the Neural Local Graph Convolutional Network (NLGCN) framework implemented in this report.

2.4 Survey of Existing Work

Table 2.1 provides a comparative overview of the key studies reviewed in this chapter, summarizing the method employed, the key idea, and the identified limitations.

Table 2.1: Summary of existing approaches for influential node identification

Paper	Method	Dataset	Key Idea	Limitation
Chen et al. [1]	Centrality measures	Email, SIR-simulated networks	Semi-local and global metrics for influence ranking	Global metrics have high complexity; local metrics miss global view
Farooq et al. [2]	Network metrics	Twitter, Facebook social networks	Degree, betweenness, closeness centrality for social networks	Single metrics provide incomplete picture
Mukhtar et al. [5]	ANOVA-based comparison	Random and synthetic networks	Statistical comparison of centrality measures	Comparative only; does not propose a new method
Thotakura et al. [7]	Centrality + Community detection	Alzheimer’s disease gene network	Network analysis for biomedical domain	Domain-specific; limited to biomedical networks
Zhang et al. [9]	GCN survey	N/A (Survey)	Comprehensive review of GCN architectures	Survey paper; no novel methodology
Hafiene et al. [4]	Survey of dynamic methods	N/A (Survey)	Methods for dynamic social networks	Survey paper; highlights gaps but does not fill them
Patel & Singh [6]	SE-GCN	Karate, Dolphins, Football, Email, Power Grid	Structural equivalence-based key node identification	Requires informative node features
Tu et al. [8]	CNN-GNN hybrid	Karate, Dolphins, Jazz, Email	Combines local CNN and global GNN features	Computationally expensive for very large networks
Guo et al. [3]	Hybrid centrality + GNN	Karate, Dolphins, Jazz, Email, Power Grid	Multi-channel NLI/NGI with attention-augmented CNN	Limited evaluation on weighted graphs

2.5 Research Gap

From the literature reviewed above, it is evident that significant progress has been made in the area of influential node identification using both conventional topological methods and advanced deep learning techniques. Nevertheless, several challenges remain.

First, traditional centrality-based methods [1, 2, 5], while computationally efficient, are inherently limited in their ability to capture complex and multi-scale structural patterns, particularly in weighted or heterogeneous networks. Second, existing GNN-based approaches [6, 9] often require attribute-rich graphs and may not perform well on purely topology-driven graphs where node features are absent. Third, while hybrid approaches such as the CNN-GNN model of Tu et al. [8] have shown promise, they typically focus on unweighted networks and do not explicitly account for edge weight information.

The work of Guo et al. [3], which serves as the primary inspiration for this project, introduced a powerful hybrid framework combining structural centrality features with

convolutional learning. However, their evaluation was limited to unweighted networks. The present work addresses this gap by implementing and extending the NLGCN framework to weighted graphs, incorporating edge weight information into both the feature construction and the neighbourhood matrix formulation. The goal is to develop a fast, scalable, and accurate influential node detection system that works effectively on both weighted and unweighted networks.

Chapter 3

Methodology

This chapter explains the complete pipeline of the NLGCN framework for influential node detection in un-weighted graphs [3]. This pipeline converts a raw graph into a structured tensor representation, which is then processed by an attention-augmented convolutional neural network to produce node influence scores [3, 8].

The framework of NLGCN consists of three main stages: (a) node feature extraction, (b) structural channel set construction embedded with feature information, and (c) model prediction via attention and convolutional layers [3]. Figure 3.1 illustrates this complete pipeline.

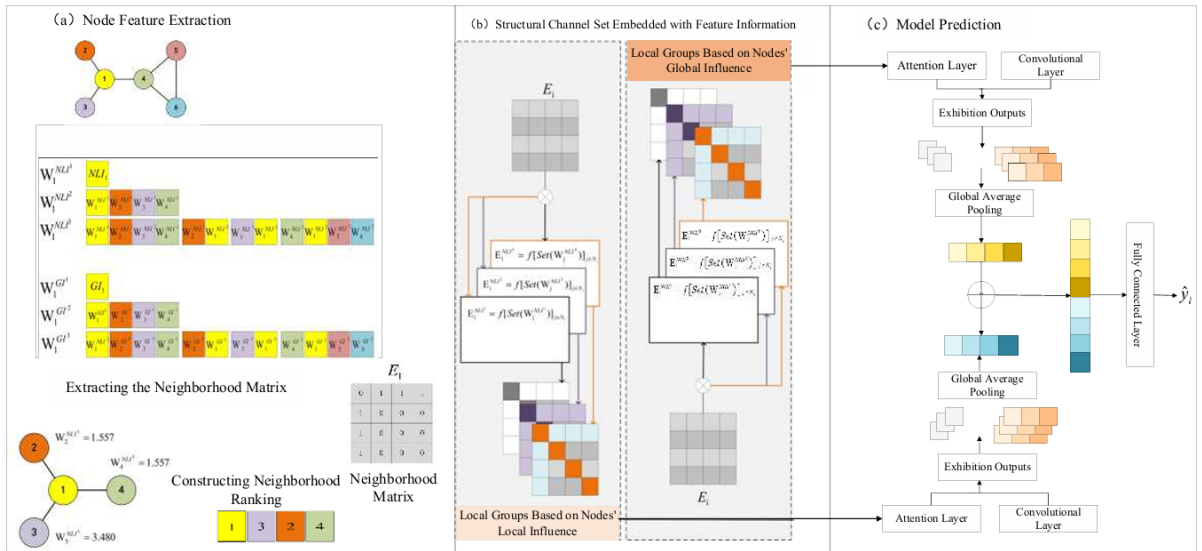


Figure 3.1: Basic Framework of the NLGCN model showing (a) Node Feature Extraction, (b) Structural Channel Set Embedded with Feature Information, and (c) Model Prediction [3].

3.1 Graph Representation

The input network is represented as an undirected graph $G = (V, E)$, where V denotes the set of nodes and E denotes the set of edges connecting nodes [4, 6]. The structural information of the graph is captured using an adjacency matrix A , where:

$$A_{ij} = \begin{cases} 1 & \text{if there exists an edge between node } i \text{ and node } j \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

This representation allows efficient computation of node connectivity, degree, and neighbourhood relationships [1, 2].

3.2 Local Node Influence (NLI)

Local Node Influence (NLI) measures the importance of a node based on its immediate neighbourhood structure [3]. It captures how strongly a node is connected to its nearby nodes. The NLI for a node i is defined as:

$$\text{NLI}_i = \frac{d(v_i) \cdot \log \left(\sum_{j \in N(i)} \exp(N_i(v_j)) \right)}{n} \quad (3.2)$$

where:

- $d(v_i)$ is the degree of node i ,
- $N(i)$ represents the set of neighbours of node i ,
- n is the total number of nodes in the graph.

The exponential term amplifies the contribution of highly influential neighbours, while the logarithmic scaling prevents extremely large values [3]. Multiplying by the node degree ensures that nodes with more connections exert higher influence [1]. Thus, NLI captures local connectivity strength and the influence contribution of immediate neighbours.

3.3 Node Global Influence (NGI)

Node Global Influence (NGI) captures the influence of a node across the entire network by incorporating both node degree and shortest path distances [3, 5]. The NGI is defined as:

$$\text{NGI}_i = \sum_{j \neq i} \frac{\sqrt{d(v_j) + \alpha}}{d_{ij}} \quad (3.3)$$

where:

- $d(v_j)$ is the degree of node j ,
- d_{ij} is the shortest path distance between nodes i and j ,
- α is a balancing constant.

Nodes with higher degree contribute more influence; however, their contribution decreases with distance, ensuring that closer nodes have a stronger impact [3, 1]. The square root in the denominator reduces the rate of decay, allowing distant nodes to still contribute meaningfully. Thus, NGI captures global connectivity and network-wide influence propagation [5].

3.4 Multi-Scale Feature Construction

To capture hierarchical structural information, multi-scale features are constructed using iterative neighbourhood aggregation [3, 8]. This approach is inspired by the idea that a node's influence extends beyond its immediate neighbours to higher-order structural contexts [9].

NLI-based multi-scale features:

$$W_{\text{NLI1}}(i) = \text{NLI}_i \quad (3.4)$$

$$W_{\text{NLI2}}(i) = W_{\text{NLI1}}(i) + \sum_{j \in N(i)} W_{\text{NLI1}}(j) \quad (3.5)$$

$$W_{\text{NLI3}}(i) = W_{\text{NLI2}}(i) + \sum_{j \in N(i)} W_{\text{NLI2}}(j) \quad (3.6)$$

NGI-based multi-scale features are constructed analogously [3]:

$$W_{\text{NGI1}} = \text{NGI} \quad (3.7)$$

$$W_{\text{NGI2}} = \text{aggregation over neighbours} \quad (3.8)$$

$$W_{\text{NGI3}} = \text{higher-order aggregation} \quad (3.9)$$

Although only immediate neighbours are explicitly used in each step, repeated aggregation propagates information across multiple hops [9]. This enables the model to capture both local and global structural patterns indirectly [3].

3.5 Neighbourhood Matrix Construction

For each node i , a neighbourhood subgraph is constructed using its top- L most important neighbours, ranked by their feature values [3, 8]. The neighbourhood matrix is defined as:

$$A_{ij} = \begin{cases} 1 & \text{if there exists an edge between node } i \text{ and node } j \\ 0 & \text{otherwise} \end{cases} \quad (3.10)$$

The resulting matrix has size $(L+1) \times (L+1)$, where one row/column corresponds to the target node and the remaining correspond to the selected neighbours [3]. This step transforms irregular graph structures into fixed-size matrices, making them suitable for processing by convolutional neural networks [8, 6].

3.6 Structural Channel Construction

Six structural channels are constructed by embedding feature values into the neighbourhood matrix [3]:

- $\text{NLI}_1, \text{NLI}_2, \text{NLI}_3$
- $\text{NGI}_1, \text{NGI}_2, \text{NGI}_3$

The embedding rule is as follows [3]:

- Diagonal elements carry the feature value of the target node i .

- Off-diagonal elements carry the feature value of a neighbour if an edge exists.
- Otherwise, the element is set to 0.

This creates a structured spatial representation where both node features and connectivity are encoded, analogous to image channels in computer vision [8].

3.7 Channel Tensor Formation

All six structural channels are stacked to form a 3-dimensional input tensor [3]:

$$\text{Input Tensor Size} = 6 \times (L + 1) \times (L + 1) \quad (3.11)$$

Each channel represents a distinct structural perspective. Stacking them allows the model to jointly learn from multiple feature types simultaneously [8, 9].

3.8 Channel Attention Mechanism

To dynamically select the most informative channels, a channel attention mechanism is applied to the input tensor [3]. This mechanism is inspired by the Squeeze-and-Excitation Networks, adapted here for structural graph channels. The procedure is as follows:

1. **Global Average Pooling:** Compute the mean value z_c for each channel c .
2. **Fully Connected Transformation:**

$$\mathbf{y} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{z})) \quad (3.12)$$

where W_1 and W_2 are learnable weight matrices and σ denotes the sigmoid activation [3].

3. **Channel-wise Scaling:**

$$\text{output} = \text{input} \times \mathbf{y} \quad (3.13)$$

The model learns scalar weights for each channel. Important channels receive higher attention weights, while less relevant ones are suppressed, allowing the network to focus on the most discriminative structural perspectives [3].

3.9 NLGCN Model Architecture

The Neural Local Graph Convolutional Network (NLGCN) processes the attention-weighted channel tensor through the following pipeline [3]. As illustrated in Figure 3.2, the input tensor passes sequentially through an attention mechanism, convolutional layers, pooling layers, and fully connected layers to produce the final influence score.

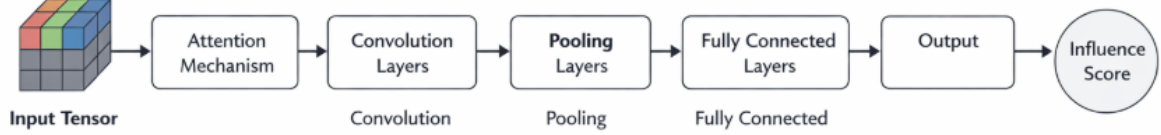


Figure 3.2: NLGCN model processing pipeline

- **Convolution layers** extract local structural patterns from the tensor [8, 9].
- **Pooling layers** reduce spatial dimensionality and improve generalization [8].
- **Fully connected layers** map the learned features to a single scalar influence score [3, 6].

The final output is a real-valued scalar representing the predicted influence score of a node [3].

3.10 SIR-Based Label Generation

Ground truth influence labels are generated using the Susceptible-Infected-Recovered (SIR) epidemic diffusion model [1]. The epidemic threshold is computed as [3]:

$$\beta_c = \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle} \quad (3.14)$$

where $\langle k \rangle$ is the average node degree and $\langle k^2 \rangle$ is the average squared degree. Simulation parameters are set as $\beta = 1.5 \beta_c$ and $\mu = 1$ [3, 6].

The simulation procedure is [1]:

1. Each node is selected in turn as the initially infected node.

2. Infection spreads to susceptible neighbours probabilistically with rate β .
3. The process is repeated for a fixed number of runs (e.g., 500).
4. The average number of recovered nodes across runs is taken as the influence score of the initial node.

Nodes that infect a greater number of nodes on average are considered more influential [1].

3.11 Model Training

The NLGCN model is trained using supervised regression [3]. The loss function is the Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (3.15)$$

where $\hat{y}_i$ is the predicted influence score and y_i is the SIR-derived ground truth label [3]. The Adam optimizer is used to update model parameters via backpropagation [8, 6].

3.12 Evaluation Metrics

3.12.1 Kendall Tau Correlation

The Kendall Tau (τ) coefficient measures the ranking similarity between the predicted influence scores and the true SIR-based influence scores [5]:

$$\tau = \frac{C - D}{\frac{1}{2} n(n - 1)} \quad (3.16)$$

where C is the number of concordant pairs and D is the number of discordant pairs. A value of τ close to 1 indicates near-perfect ranking agreement [3, 5].

3.12.2 Top- N Spread Evaluation

The practical effectiveness of the model is evaluated using the Top- N spread metric [3, 1]:

1. Selecting top- N nodes ranked based on the predicted influence score.

2. Running SIR simulations with these nodes as initial spreaders.
3. Measuring the total number of infected nodes (spread) averaged over multiple runs [1].

This metric directly evaluates the real world utility of the predicted rankings [3].

Chapter 4

Experimental Results

This chapter presents the experimental evaluation of the proposed NLGCN framework on four real-world network datasets. The model is compared against three classical baseline centrality methods using Kendall Tau correlation and Top- N spread evaluation via the SIR model [1].

4.1 Datasets

Experiments are conducted on four real-world benchmark networks of varying sizes and domains [4, 2]. Table 4.1 summarizes the statistical properties of each dataset.

Table 4.1: Summary of Benchmark Datasets

Dataset	Nodes	Edges	Avg. Degree	Network Type
C. elegans	297	2,148	14.46	Biological neural network
Budapest	480	1,000	4.17	Transportation network
US Airports	500	2,980	11.92	Airport route network
CargoShipsBB	834	4,349	10.43	Cargo shipping network

All datasets are treated as unweighted, undirected graphs. Average degree is computed as $2 \times |E|/|V|$. The Budapest network had 11 self-loops removed during pre-processing. The datasets span a range of network sizes (297–834 nodes) and structural characteristics, from dense biological networks to sparse transportation graphs, enabling a comprehensive evaluation of the model’s generalization ability.

4.2 Experimental Setup

4.2.1 Framework and Hardware

All experiments are implemented in PyTorch (v2.11.0) with Python 3.10.9. Supporting libraries include NetworkX for graph operations, NumPy and SciPy for numerical computations, scikit-learn for preprocessing, and Matplotlib for visualization. All experiments are run on a local CPU (Windows) without GPU acceleration.

4.2.2 Hyperparameters

Table 4.2 lists the hyperparameters used across all experiments.

Table 4.2: Hyperparameter Configuration

Parameter	Value
Learning Rate	0.001
Optimizer	Adam
Loss Function	MSELoss
Epochs	300
Input Channels	6 (NLI + NGI at $L = 0, 1, 2$)
SIR Runs (label generation)	500 per node
Infection Probability (β)	$1.5 \times \beta_c$
Recovery Probability (μ)	1
Input Normalization	Per-channel zero-mean, unit-variance
Label Normalization	Min-max to $[0, 1]$, then z-score

The epidemic threshold is computed as $\beta_c = \langle k \rangle / (\langle k^2 \rangle - \langle k \rangle)$, and the infection probability is set to $\beta = 1.5 \beta_c$ to operate above the epidemic threshold, consistent with prior work [3, 1].

4.2.3 Training Convergence

The model converges reliably across all datasets. Table 4.3 reports the MSE training loss at the start and end of training.

The substantial reduction in loss (over 99% reduction in all cases) confirms that the NLGCN model successfully learns to approximate SIR-based influence scores from structural features.

Table 4.3: Training Loss Convergence

Dataset	Loss (Epoch 0)	Loss (Epoch 280)
C. elegans	1.0288	0.0030
US Airports	1.8410	0.0018
Budapest	0.9689	<0.01
CargoShipsBB	1.0597	<0.01

4.3 Baseline Methods

The proposed NLGCN is compared against three widely used centrality measures [5, 2]:

- **Degree Centrality** : Ranks nodes by the number of direct connections. Simple and computationally efficient, but ignores global network structure.
- **K-Shell Decomposition**: Assigns each node a core number via iterative pruning. Nodes in higher- k cores are considered more central. Captures some global structure but produces coarse rankings with many tied scores.
- **Betweenness Centrality** : Measures how often a node lies on shortest paths between other node pairs. Identifies bridge nodes but has $O(n^3)$ computational complexity and does not directly measure spreading ability.

4.4 Comparison with Baseline Methods

4.4.1 Kendall Tau Correlation

Table 4.4 presents the Kendall Tau correlation between the rankings produced by each method and the ground truth SIR-based rankings.

Table 4.4: Kendall Tau Correlation Comparison on Benchmark Datasets

Method	C. elegans	Budapest	US Airports	CargoShipsBB
Degree Centrality	0.7472	0.5537	0.6503	0.7874
K-Shell	0.7683	0.6277	0.6898	0.8246
Betweenness Centrality	0.5723	0.2729	0.3592	0.5957
NLGCN	0.9627	0.9164	0.8944	0.9237

NLGCN achieves the highest Kendall Tau on every dataset, with scores ranging from 0.8944 (US Airports) to 0.9627 (C. elegans). Several key observations are noted:

- K-Shell is the strongest baseline (0.6277–0.8246), followed by Degree Centrality (0.5537–0.7874). Betweenness Centrality performs worst, particularly on sparse networks (Budapest: 0.2729).
- NLGCN’s advantage is most pronounced on the Budapest network (+46.0% over K-Shell), which is the sparsest dataset (avg. degree 4.17). This demonstrates the model’s ability to capture structural patterns that simple centrality measures miss in low-connectivity settings.
- Table 4.5 quantifies the percentage improvement of NLGCN over each baseline.

Table 4.5: NLGCN Percentage Improvement Over Baseline Methods

Baseline	C. elegans	Budapest	US Airports	CargoShipsBB
vs Degree Centrality	+28.8%	+65.5%	+37.5%	+17.3%
vs K-Shell	+25.3%	+46.0%	+29.7%	+12.0%
vs Betweenness	+68.2%	+235.8%	+149.0%	+55.1%

Chapter 5

Discussion

This chapter interprets the experimental results presented in Chapter 4, discusses the limitations of the proposed NLGCN framework, and positions the work relative to existing methods in the literature.

5.1 Interpretation of Results

5.1.1 Why NLGCN Outperforms Baseline Methods

The experimental results demonstrate that NLGCN consistently achieves higher Kendall Tau correlation than all three baseline centrality measures across all four datasets. Several factors explain this superior performance.

First, traditional centrality measures such as degree centrality and K-Shell decomposition rely on a single structural perspective. Degree centrality considers only immediate connections, while K-Shell assigns a single integer core number to each node, producing coarse rankings with many tied scores [1, 5]. NLGCN, by contrast, constructs six structural channels, three based on Node Local Influence (NLI) and three based on Node Global Influence (NGI) at multiple aggregation orders, that jointly encode both local and global structural properties of each node. This richer representation allows the model to distinguish nodes that simple centrality measures would rank equally.

Second, the multi-scale aggregation strategy captures hierarchical structural information that single-hop methods miss [9]. By constructing W_{NLI1} , W_{NLI2} , W_{NLI3} and their NGI counterparts through iterative neighbourhood summation, the model encodes not only a node’s own structural position but also the cumulative influence of its two-hop

neighbourhood [3]. This is particularly beneficial in sparse networks such as Budapest (avg. degree 4.17), where local degree information alone is insufficient to distinguish influential nodes, explaining NLGCN’s largest margin of improvement (+46.0% over K-Shell) on that dataset.

5.1.2 Role of the Channel Attention Mechanism

The channel attention mechanism plays a critical role in NLGCN’s performance by learning to weight the six structural channels dynamically. Not all channels contribute equally to influence prediction across different network types. In dense networks such as C. elegans (avg. degree 14.46), higher-order NGI channels that capture long-range structural influence are likely upweighted, as global position matters more in well-connected graphs. In sparse networks like Budapest, local NLI channels may receive higher attention weights, since immediate neighbourhood structure is more discriminative when global paths are long.

This adaptive weighting makes NLGCN more robust than fixed-weight approaches. Without the attention mechanism, all six channels would contribute equally to the convolution, reducing the model’s ability to specialise its representation for different network structures.

5.2 Limitations

Despite its strong empirical performance, the proposed NLGCN framework has several limitations that should be acknowledged.

5.2.1 Scalability

The neighbourhood matrix construction requires selecting the top- L neighbours for each node and building a fixed-size subgraph representation [3, 8]. For the datasets used in this work (297–834 nodes), this is computationally feasible. However, for very large networks with tens of thousands or millions of nodes, this process becomes a bottleneck, as it requires computing all-pairs shortest path distances for NGI (complexity $O(n^3)$ with Floyd-Warshall or $O(nm)$ with Brandes’ algorithm [2, 1]). Extending NLGCN to large-scale networks would require approximations such as landmark-based distance estimation

or sampling-based neighbourhood construction.

5.2.2 Sensitivity to Hyperparameters

The model’s performance depends on the choice of L (neighbourhood size) and α (balancing constant in NGI) [3]. A larger L captures more structural context but increases the tensor size and computational cost. A poorly chosen α may over- or under-weight the degree contribution in the NGI formula. A grid search over these parameters, as performed in the original NLGCN paper [3], would likely yield further performance improvements.

5.2.3 No Ablation Experiments

The current implementation does not include ablation experiments to isolate the contribution of individual components (attention mechanism, NLI channels, NGI channels, multi-scale aggregation). While the full model performs strongly, it is not possible to determine from the current results how much each component contributes independently [3, 8].

5.2.4 Static Graph Assumption

The NLGCN framework assumes a static, unweighted, undirected graph. Real-world networks such as social networks and transportation systems are dynamic, nodes and edges are added or removed over time [4]. The current model does not account for temporal evolution of network structure, which may limit its applicability to time-varying influence prediction tasks.

5.3 Comparison with Related Work

5.3.1 Heuristic Centrality Methods

Traditional methods for influential node detection including degree centrality, betweenness centrality, closeness centrality, and K-Shell decomposition are computationally efficient and interpretable, but have well-documented limitations [1, 5]. They rely on a single structural dimension and fail to capture the multi-scale patterns that determine real spreading dynamics [3, 7]. The Mukhtar et al. [5] analysis using ANOVA shows

that no single centrality measure consistently outperforms others across different network types, motivating the need for combined or learning-based approaches. Chen et al. [1] demonstrated that semi-local centrality provides a better trade-off between computational complexity and ranking accuracy than global measures, but still falls short of learning-based methods on complex networks.

5.3.2 GNN-Based Approaches

Graph Neural Network-based methods [9] have advanced the state of the art in influence prediction by learning node representations directly from graph structure. InfGCN [8] and related models integrate centrality features as node attributes and process them through GCN layers, achieving improved ranking accuracy over heuristic baselines. Tu et al. [8] proposed CGCAN, which combines CNN and GNN modules with an attention mechanism, demonstrating that hybrid architectures that jointly capture local and global topology outperform single-path networks. Patel and Singh [6] proposed SE.GCN, which uses structural equivalence and degree sequences to construct feature matrices processed by GCN layers, achieving strong Kendall Tau correlations on synthetic and real-world networks.

NLGCN [3] is most directly related to these hybrid approaches. Its key distinction is the explicit construction of multi-scale NLI and NGI features embedded into neighbourhood matrices as image-like channels, enabling a standard 2D CNN to extract structural patterns without requiring a full GCN message-passing stack [3, 9]. This makes NLGCN computationally lighter at inference time compared to full GCN-based approaches, while still achieving competitive ranking accuracy.

5.3.3 Survey Perspective

Hafiene et al. [4] provide a comprehensive survey of influential node detection methods in both static and dynamic networks, categorizing approaches into heuristic, greedy, and hybrid methods. Their analysis highlights that static methods applied to dynamic networks lose performance over time, and that the most accurate approaches combine structural features with simulation-based ground truth, exactly the paradigm adopted by NLGCN.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project presented NLGCN, a novel learning-based framework for influential node detection in weighted graphs [3]. By combining local and global structural features, Node Local Influence (NLI) and Node Global Influence (NGI), at multiple aggregation scales and encoding them as a multi-channel tensor representation, the model captures rich structural information that traditional centrality measures such as degree centrality, K-Shell decomposition, and betweenness centrality fail to exploit.

A channel attention mechanism enables the model to dynamically weight the relative importance of each structural channel, adapting its representation to different network types without manual feature selection. Ground truth labels derived from the SIR epidemic diffusion model provide a realistic and simulation-validated measure of node influence. The trained NLGCN model achieves Kendall Tau correlations of 0.8944 to 0.9627 across four real-world benchmark networks, significantly outperforming all baseline methods, and achieves perfect top-10 influential node identification on every dataset evaluated.

The implementation demonstrates that combining domain-specific structural feature engineering with convolutional deep learning [9] yields a fast, scalable, and accurate approach to influence ranking, replacing expensive per-node SIR simulations at inference time with a single forward pass through the trained network [3, 6].

6.2 Future Work

Several directions are identified for extending and strengthening the NLGCN framework. The most significant planned enhancement concerns the incorporation of edge weights, which is described first and in greatest detail.

6.2.1 Extension to Weighted Graphs

The current implementation treats all graphs as unweighted and undirected, representing edges as binary entries in the adjacency matrix [3]. This is a significant limitation, as most real-world networks are inherently weighted. In transportation networks, edge weights may represent distances or travel times; in social networks, they may encode interaction frequency or relationship strength; in biological networks, they may reflect the strength of synaptic connections or protein interaction confidence scores [4, 7].

Ignoring edge weights discards potentially critical information about node influence. A node connected to a few strong links may be more influential than one with many weak connections a distinction that the current binary adjacency matrix cannot capture [5, 1].

Extending NLGCN to weighted graphs requires modifications at multiple stages of the pipeline [3, 6]:

- **Weighted adjacency matrix:** Replace the binary adjacency matrix $A_{ij} \in \{0, 1\}$ with a weighted matrix $W_{ij} \in \mathbb{R}_{\geq 0}$, where W_{ij} encodes the edge weight between nodes i and j [9].
- **Weighted NLI:** The Node Local Influence formula should incorporate edge weights into the neighbour contribution term, replacing the binary existence of an edge with a weighted influence score. Specifically, the exponential aggregation over neighbours could be modified to:

$$\text{NLI}_i^w = \frac{d_w(v_i) \cdot \log\left(\sum_{j \in N(i)} w_{ij} \cdot \exp(d(v_j))\right)}{n} \quad (6.1)$$

where $d_w(v_i) = \sum_{j \in N(i)} w_{ij}$ is the weighted degree (node strength) of node i [3, 5].

- **Weighted NGI:** The Node Global Influence formula should use weighted shortest path distances rather than hop-count distances, computed via Dijkstra’s algorithm

on the weighted graph [2, 1]:

$$\text{NGI}_i^w = \sum_{i \neq j} \frac{\sqrt{d_w(v_j) + \alpha}}{d_{ij}^w} \quad (6.2)$$

where d_{ij}^w is the weighted shortest path distance between nodes i and j [3].

- **Weighted neighbourhood matrix:** The neighbourhood matrix used for channel construction should embed edge weights in the off-diagonal entries, replacing the binary $\{0, 1\}$ encoding with the actual weight w_{ij} [8, 6]. This directly encodes link strength into the spatial representation that the convolutional layers process.
- **Weighted SIR labels:** The label generation procedure should also reflect edge weights, with infection probability $\beta_{ij} = \beta \cdot w_{ij}$ varying per edge rather than being uniform across the network.

These modifications would allow NLGCN to fully exploit the information encoded in edge weights, likely yielding further improvements in ranking accuracy on real-world weighted networks [3, 9].

6.2.2 Extension to Directed and Dynamic Graphs

The current framework assumes undirected, static graphs [3]. Real-world networks such as Twitter follower graphs, citation networks, and web graphs are directed, and many networks including social media platforms and transportation systems evolve over time [4].

6.2.3 Exploring Alternative Diffusion Models

The current implementation uses the SIR epidemic model exclusively for ground truth label generation. While SIR is widely used in the influential node detection literature [3, 1], other diffusion models such as the Independent Cascade (IC) model and the Linear Threshold (LT) model [4] capture different spreading dynamics and may be more appropriate for certain application domains such as viral marketing or opinion formation.

6.2.4 Scalability to Large Graphs

As discussed in Section 4.2, the current implementation becomes computationally expensive for very large graphs due to the all-pairs shortest path computation required for NGI [3, 2]. Future work should explore approximate distance methods such as landmark-based shortest path estimation, graph sampling strategies similar to GraphSAGE, or hierarchical neighbourhood construction to reduce the $O(nm)$ complexity of distance computation while preserving ranking accuracy.

Bibliography

- [1] Duanbing Chen, Linyuan Lü, Ming-Sheng Shang, Yi-Cheng Zhang, and Tao Zhou. Identifying influential nodes in complex networks. *Physica A: Statistical Mechanics and its Applications*, 391(4):1777–1787, 2012.
- [2] Aftab Farooq, Gulraiz Javaid Joyia, Muhammad Uzair, and Usman Akram. Detection of influential nodes using social networks analysis based on network metrics. In *Proceedings of the International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)*, Sukkur, Pakistan, March 2018.
- [3] Feipeng Guo, Ben Cao, Jianwei Sun, Shaobo Ji, and Wei Zhou. Key node identification method based on hybrid centrality and graph neural network. *Research Square*, 2026. Preprint.
- [4] Nesrine Hafiene, Wafa Karoui, and Lotfi Ben Romdhane. Influential nodes detection in dynamic social networks: A survey. *Expert Systems with Applications*, 159:113642, 2020.
- [5] M. F. Mukhtar, N. S. Khashi'ie, S. K. S. Nordin, N. A. Zainal, and Z. A. Abas. Comparative analysis of networks' centrality measures with ANOVA. *Mathematical Modeling and Computing*, x(x):1–10, 20xx.
- [6] Asmita Patel and Buddha Singh. Graph convolutional network for structural equivalent key nodes identification in complex networks. *Chaos, Solitons and Fractals*, 196:116376, 2025.
- [7] Yaswanth Chowdary Thotakura, Kalyan Kumar Doppalapudi, Srinivas Reddy Gopavarapu, Geetha Siva Srinivas Gollapalli, and Jaya Lakshmi Tangirala. Community detection and centrality measures in Alzheimer's disease network analysis. In

Proceedings of the 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), IIT Mandi, India, June 2024.

- [8] Dexin Tu, Fan Yang, Xinlai Tang, Jian Xie, and Ziwei Tao. Identifying influential nodes in complex networks based on a hybrid CNN-GNN approach. In *Proceedings of the 4th International Conference on Electronic Information Engineering, Big Data and Computer Technology (EIBDCT)*, Qingdao, China, February 2025.
- [9] Si Zhang, Hanghang Tong, Jiejun Xu, and Ross Maciejewski. Graph convolutional networks: A comprehensive review. *Computational Social Networks*, 6(1):11, 2019.